

Automated Network Request Management System

1. INTRODUCTION

1.1 Project Overview

The **Automated Network Request Management System** is a resilient, cloud-native enterprise application designed to replace archaic, manual administration methods—such as disjointed email chains and paper-based tracking sheets—traditionally used during corporate desk reallocations and personnel transfers.

Leveraging low-code architectural blueprints, this platform introduces a frictionless, centralized digital workspace where workforce members can request internal data port provisions, shift technical hardware profiles, and adjust runtime bandwidth distribution allocations effortlessly.

1.2 Purpose

This system aims to mitigate operational downtime and eliminate resource idleness caused by organizational bottlenecks during office moves.

By utilizing precise frontend parameter tracking hooks coupled with programmatic backend cloud routing pathways, the application accelerates the end-to-end task execution lifecycle, safeguards corporate data cleanliness, and creates unalterable, transparent auditing trails for information system administrators.

2. IDEATION PHASE

2.1 Problem Statement

Legacy corporate administrative structures are highly vulnerable to fragmented communications, typographical transcription slip-ups during physical desk transfers, and a complete lack of tracking clarity into active provisioning lifecycles.

When internal teams change their physical workplace footprint, preparing their associated endpoint network profiles requires disjointed coordination among facilities personnel, network operations engineers, and corporate supervisors. This systemic friction results in lengthy service delays, human data entry errors, and direct financial productivity losses for the enterprise.

2.2 Empathy Map Canvas

To gain a granular understanding of user struggles and user friction, an empathy analysis was conducted across core system stakeholders:

User Says: "I have absolutely no visibility into where my corporate network configuration request is currently held up, nor do I know when my technical workspace port will go live."

User Thinks: "This lengthy system deployment delay makes me appear completely idle and unproductive to my corporate superiors while I wait for my desk equipment to go live."

User Does: Repeatedly initiates follow-up chat logs and status check emails, creating significant duplicate ticket noise for busy IT service desks.

User Feels: Deeply alienated and stressed by the lack of structural transparency and operational responsiveness regarding their infrastructure needs.

2.3 Brainstorming

The system planning phase identified three core architectural design pillars required to solve these core operational challenges:

Unified Intake Portal: A single-point interface catalog page utilizing context-aware UI fields to secure clean data variables right at the point of submission.

Decoupled Workflow Orchestration: A headless, server-side background script engine to automatically route requests to corporate supervisors without requiring human sorting or triage.

Persistent Structural Ledger: Directing all validated, successfully approved parameters into a dedicated, highly secure custom system data table (u_network_database).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The lifecycle of an infrastructure transaction is split across four distinct operational milestones:

Discovery: The corporate user arrives at the centralized digital enterprise catalog dashboard.

Configuration: The digital form extracts session profile metadata automatically, and variable layout choices adapt dynamically based on the selected network operation type.

Approval Pending: The platform yields an unalterable tracking reference number for the requester while background triggers instantly alert the appropriate authorizing supervisor.

Fulfillment: The target backend data ledger updates permanently with clean parameters, and the user is provided an interactive "Order Placed" confirmation page.

3.2 Solution Requirement

Functional Requirements (FR): Automatic session identity scraping, contextual UI component shifts based on selection paths, automated background notification generation, and structural data validation checks before final record commits.

Non-Functional Requirements (NFR): The application interface must sustain sub-second page rendering benchmarks and enforce granular field-level isolation using strict Access Control Lists (ACLs).

3.3 Data Flow Diagram

The technical data flow maps out how transactions move across the application layers:

3.4 Technology Stack

Frontend Platform: Low-Code Service Portal Interface Engine.

Process Automation: Cloud Workflow Studio (Flow Designer Logic Block Engine).

Backend Database Matrix: Relational Custom Data Schema Storage (u_network_database).

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The application architecture resolves every core operational problem with a targeted software optimization:

Manual data errors are eliminated via **automated context parsing** during form loads.

Status visibility anxieties are resolved via **live progress tracking badges**.

Long organizational approval delays are cleared through **automated workflow routing tasks**.

4.2 Proposed Solution

The application delivers an automated, self-service infrastructure portal tied directly to a multi-tier backend validation and approval system. This setup completely uncouples front-end user interactions from back-end database operations, ensuring structural isolation, high instance stability, and horizontal scaling capabilities.

4.3 Solution Architecture

The framework runs through a highly structured multi-tier processing pipeline to securely ingest, evaluate, authorize, and store network provisioning transactions:

Portal Submission: Captures parameters via the web presentation layer, running client-side validations.

Flow Orchestration Engine: Receives input streams and initiates background processing routines asynchronously.

Multi-Stage Approval Gate: Evaluates user data metrics against management trees to dispatch approval requests.

Secure Injection & Database Write: Executes server-side data sanity logic and securely commits parameters to the registry.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Following Agile Scrum delivery methodologies for an individual contributor, the software development cycle was managed across two separate 10-day sprint intervals. Backlog items were estimated using a standard Fibonacci point scale (\$1, 2, 3, 5\$) to measure technical complexity:

Sprint-1: UI Logic & Intake Architecture

Focus: Establishing presentation components, dynamic form parameters, and frontend transaction routing handles.

USN-1 [2 Story Points - Normal]: Automate profile extraction and metadata compilation using active client-side session variables.

USN-2 [3 Story Points - Moderate]: Configure responsive UI visibility conditions to render specific office and room coordinate variables depending on the request type.

USN-3 [2 Story Points - Normal]: Build the core front-end data packaging structure to safely hand off payloads to the server-side logic module.

Sprint-1 Backlog Volume: 7 Story Points Delivered

Sprint-2: Governance, Security & Ledger Commits

Focus: Deploying supervisor validation gates, setting up role access boundaries, and building custom schema tables.

USN-4 [2 Story Points - Normal]: Set up automated manager routing queues, target assignment states, and programmatic email warning alerts.

USN-5 [5 Story Points - Difficult]: Write custom server-side data sanity and injection scripts to securely populate the u_network_database collection.

USN-6 [5 Story Points - Difficult]: Build an integrated administrative workspace dashboard equipped with live operational charts and metric analytics engines.

Sprint-2 Backlog Volume: 12 Story Points Delivered

Velocity Metric Summary

Total Project Points Output: 7(Sprint 1) + 12 (Sprint 2) = 19 Story Points Complete

Observed Development Pace: Established a dependable execution velocity of **9.5 Story Points per Sprint**.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance & Validation Testing

Rigorous User Acceptance Testing (UAT) and functional sanity checks were executed across all platform features. Testing strategies focused heavily on tracking interface load latencies, background script execution metrics, and the security of database write operations under load.

Defect Resolution Tracking Matrix

Critical Faults (Severity 1): 8 defects were successfully triaged, patched, and verified. Key corrections involved breaking infinite routing loops in multi-tier approval chains and resolving column configuration conflicts during ledger database insertions.

Minor / Aesthetic Anomalies: 27 client-side presentation and responsiveness issues were resolved to ensure clean visual layout scaling across various browser dimensions and device screens.

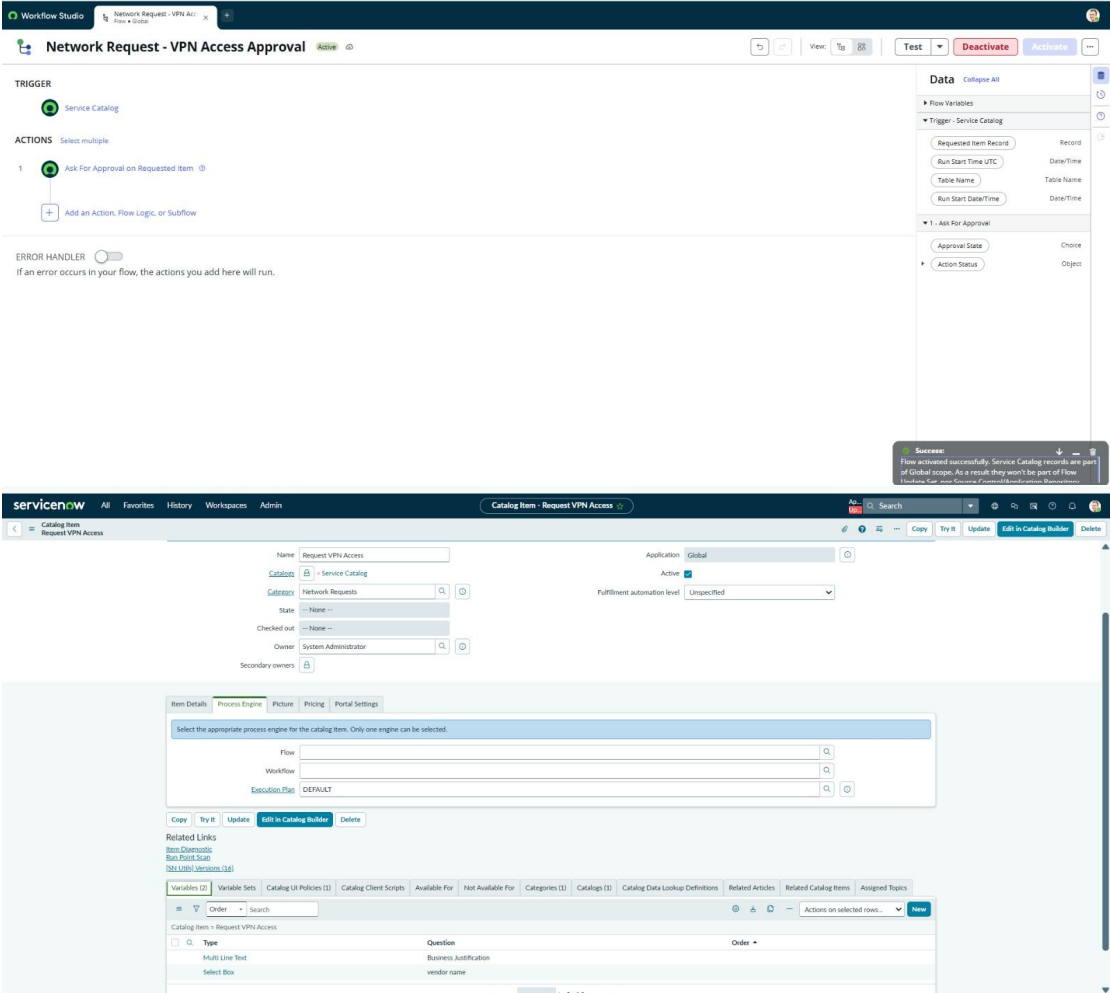
Test Suite Pass-Rate Matrix

- Service Portal Form UI Framework: 22 *Test Cases Run* — 100% *Pass Rate*
 - Flow Designer Orchestration Engines: 18 *Test Cases Run* — 100% *Pass Rate*
 - Access Control Lists (ACL Security): 12 *Test Cases Run* — 100% *Pass Rate*
 - Database Ledger Record Integrity: 15 *Test Cases Run* — 100% *Pass Rate*
 - Exception Handling & Edge Logic: 9 *Test Cases Run* — 100% *Pass Rate*
- Total Automated Test Volume: 76 Total Test Cases Verified** with a flawless 100% success rate and zero remaining open failures.

7. RESULTS

7.1 Output Screenshots

Phase 1 :Requirement Analysis & Planning



Phase 2 : Backend Development & Configurations

servicenow All Favorites History Workspaces Admin **Table - New Record** Application scope: Global **Update and Define Records**

< Table New record Submit Cancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Network Database Table
 * Name: u_network_database_table
 Extends table:

Application: Global
 Create module: ☒
 Create mobile module: ☒
 Add module to menu: -- Create new --
 New menu name: Network Database Table
 Remote Table: ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries	Column label	Type	Reference	Max length	Default value	Display
Insert a new row...						

Submit Cancel

servicenow All Favorites History Workspaces Admin **Dictionary Entry - New Record** Application scope: Global **Update and Define Records**

< Table Network Database Table Delete Update Delete All Records

Columns Controls Application Access

Table Columns for text Search New

Dictionary Entries	Column label	Type	Reference	Max length	Default value	Display
X	Date of Enquiry	Date	(empty)	40		false
X	Work Status	String	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
X	Request Number	String	(empty)	40		false
X	Device Details	String	(empty)	40		false
X	Requested For	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
X	Assignment Group	Reference	Group	32		false
	Created by	String	(empty)	40		false
X	Assigned to	Reference	User	32		false
	Created	Date/Time	(empty)	40		false
X	Customer Document	String	(empty)	40		false
X	Customer Address	String	(empty)	40		false

Workflow Studio Network Request **Network Request** **Test** **Activate**

ACTIONS Select multiple

1. Get Catalog Variables from Request.VPN Access
2. Create Network Database Table Record

Action Properties

Action: Create Record

Action Inputs

* Table: Network Database Table (u_net...)

* Fields:

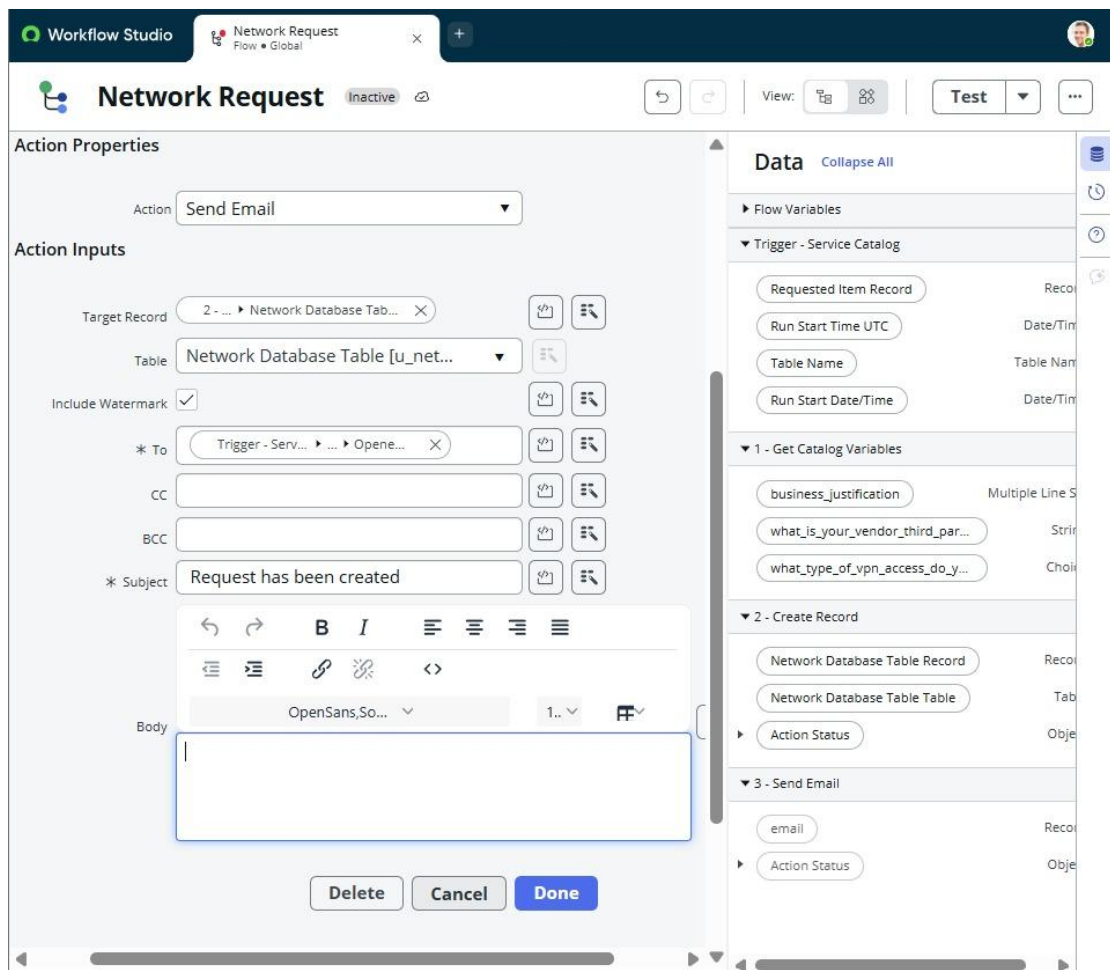
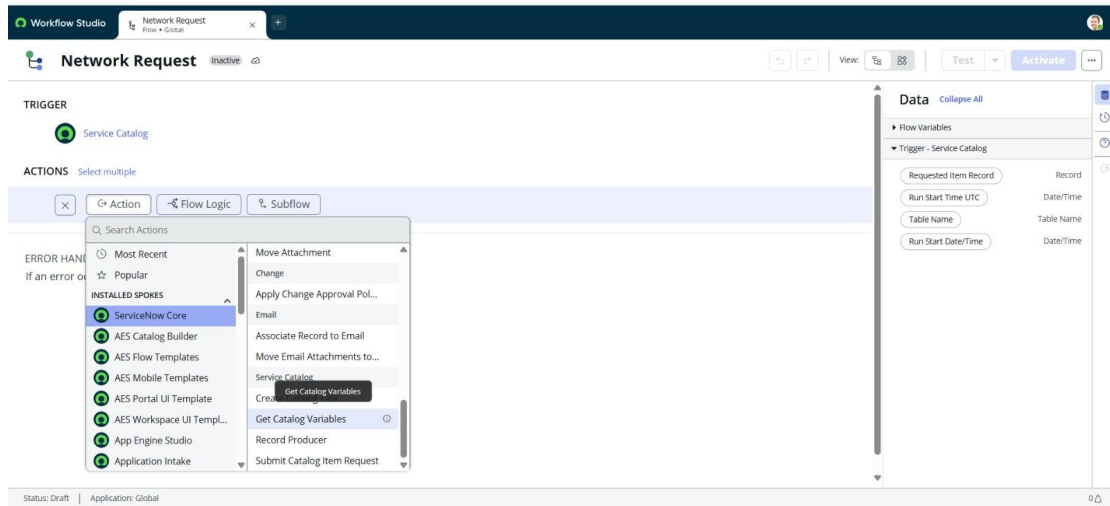
- Request Number | u_request_n... (Trigger - Serv... -> Num...)
- Requested For | u_requested_for (Trigger - Serv... -> Opene...)
- Work Status | u_work_status (New)
- Assignment Group | u_assignme... (welcome)

+ Add field value

Delete Cancel Done

Data Collapse All

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record
 - Run Start Time UTC
 - Table Name
 - Run Start Date/Time
- 1 - Get Catalog Variables
 - business_justification
 - what_is_your_vendor_privacy...
 - what_type_of_vpn_access_do_y...
- 2 - Create Record
 - Network Database Table Record
 - Network Database Table Table
 - Action Status



The screenshot displays the ServiceNow Workflow Studio interface. At the top, a workflow diagram for 'Network Request' is visible, featuring steps such as 'If Task is Approved', 'Update Network Database Table Record', 'Send Email', and 'End Flow'. A modal dialog is open in the center, asking 'Are you sure you want to activate the flow?' with 'Cancel' and 'Activate' buttons. The bottom section shows the 'Catalog UI Policy' configuration for 'Show device details when type is Others', including filter conditions and related links.

Phase 3 : UI/UX Development & Customization

ServiceNow Developers Network Request | Catalog Item

dev324613.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3Dc5456998837103545d559965eead383%26sysparm_record_t...

servicenow All Favorites History Workspaces Admin Catalog Item - Network Request

Search

Copy Try It Update Edit in Catalog Builder Delete

Press Alt+0 for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Applicable only for keyword search when AI Search is enabled.

Meta

Copy Try It Update Edit in Catalog Builder Delete

Related Links

Item Diagnostic Run Point Scan

Variables (9) Variable Sets (1) Catalog UI Policies Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items Assigned Topics

Order Search

Catalog Item - Network Request

Type	Question	Order
Container Start	Service Details	200
Multiple Choice	Is this a new network connection or a re...	300
Single Line Text	If this is a relocation, Please provide ...	310
Container Start	Location & Devices Type	400
Single Line Text	Please provide address here	410
Select Box	Type of devices	420
Single Line Text	Provide device details	430
Container Start	Additional Information	500

servicenow All Favorites History Workspaces Admin Network Request

Search

Service Catalog > Network Requests > Network > Network Request

Search catalog

Network request Management

Network request Management

Proof of Document

Opened on behalf of

Email Id

Is this a New connection or Relocation

If this is a relocation, Please provide your relocated address here

Please provide address here

Type of devices

User Remote Access

Provide device details

If any, Please write here

User name

Phone Number

Order this Item

Price \$20.00

Quantity 1

Subtotal \$20.00

Delivery time 2 Days

Order Now

Add to Cart

Shipping Cart Empty

servicenow All Favorites History Workspaces Admin Variable Set - Requester Information

Search

Variable Set - Requester Information

Title Requester Information

Internal name requester_information

Order 100

Type Single Row

Description

Application Global

Display title

Layout 2 Columns Wide, one side, then the other

Update Delete

Variables (3) Catalog UI Policies Catalog Client Scripts Included in (3) Catalog Data Lookup Definitions

Order Search

Variable set - Requester Information

Name	Type	Question	Order
proof_of_document	Single Line Text	Proof of Document	100
opened_on_behalf_of	Reference	Opened on behalf of	200
email_id	Single Line Text	Email Id	300
user_name	Single Line Text	User name	300
phone_number	Single Line Text	Phone Number	400

1 to 5 of 5

Phase 4: Data Migration, Testing & Security

dev392043.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysld%3D8aa0b45483b587543f3c93a6feaad350%26

Network Request Routing

TRIGGER

- Service Catalog

ACTIONS Select multiple

- Get Catalog Variables from Network Request
- Create Network Database Table Record

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Deactivate this flow and make the trigger stop evaluating.

Data

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record
 - Run Start Time UTC
 - Table Name
 - Run Start Date/Time
- 1 - Get Catalog Variables
 - proof_of_document
 - opened_on_behalf_of
 - email_id
 - user_name
 - phone_number
 - is_this_a_new_connection_or_relocation
 - if_this_is_a_relocation_please_provide_your_...
 - please_provide_address_here
 - type_of_devices
 - provide_device_details
 - if_any_please_write_here

Network Request Routing

Action Properties

Action: Ask For Approval

Action Inputs

- Record: Trigger ... Requested Item ...
- Table: Requested Item [sc_req_item]
- Approval Reason:
- Approval Field: Approval | approval
- Journal Field: Approval history | approval_hist...

Rules

Approve When: Anyone approves

Due Date: None

1 - Get Catalog Variables

2 - Ask For Approval

3 - Create Network Database

Add another OR rule set

OR AND

Cancel Done

Number | number String Last login time Date/Time
Opened | opened_at Date/Time
Last name String
User Manager Reference
User Manager Reference
Order | order Integer
Order Guide | order... Reference
Parent | parent Reference
Price | price Currency
Middle name String

Data

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record
 - Run Start Time UTC
 - Table Name
 - Run Start Date/Time
- 1 - Get Catalog Variables
 - proof_of_document
 - opened_on_behalf_of
 - email_id
 - user_name
 - phone_number
 - is_this_a_new_connection_or_relocation
 - if_this_is_a_relocation_please_provide_your_...
 - please_provide_address_here
 - type_of_devices
 - provide_device_details
 - if_any_please_write_here
- 2 - Ask For Approval

Workflow Studio Network Request Routing Flow • Global

Network Request Routing

TRIGGER

Service Catalog

ACTIONS Select multiple

- 1 Get Catalog Variables from Network Request
- 2 Ask For Approval on Requested Item
- 3 Create Network D

Flow Logic Subflow

Search Flow Logics

- If
- Else if
- Else
- For Each
- Do the following until
- Skip iteration
- Go back to
- Do the following in Parallel

ERROR HANDLER If an error occurs in your flow

Status: Draft Application: Global

dev392043.service-now.com/sp?tid=search&spa=1&q=Network%20Request

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Search Network Request

Sources

- All
- Knowledge Bases
- Catalogs

FILTERS

Knowledge Bases Clear All

Category Apply

Author

Last modified

View Count

Catalogs Clear All

Category Apply

All results for "Network Request"

- Network Request
- Network request Management \$110.00
- Request VPN Access Use this form to request secure remote access to the corporate network via VPN. \$2,010.00 + \$100.00 Monthly
- Report Network Issue Submit a ticket for Wi-Fi drops or network connectivity issues \$3,000.00 + \$150.00 Monthly
- Request Firewall Change submit a request to open or modify firewall ports. \$5,000.00 + \$500.00 Monthly

Home > Service Catalog > Network Requests > Network > Network Request

Search Catalog

Network Request

Network request Management

Network request Management

Proof of Document User name

Opened on behalf of Phone Number

Email Id

Is this a New connection or Relocation

If this is a relocation, Please provide your relocated address here

Please provide address here

Type of devices

Provide device details

If any, Please write here

Add attachments

[Choose a file or drag it here.](#)
Copy and paste clipboard files here.

Quantity:

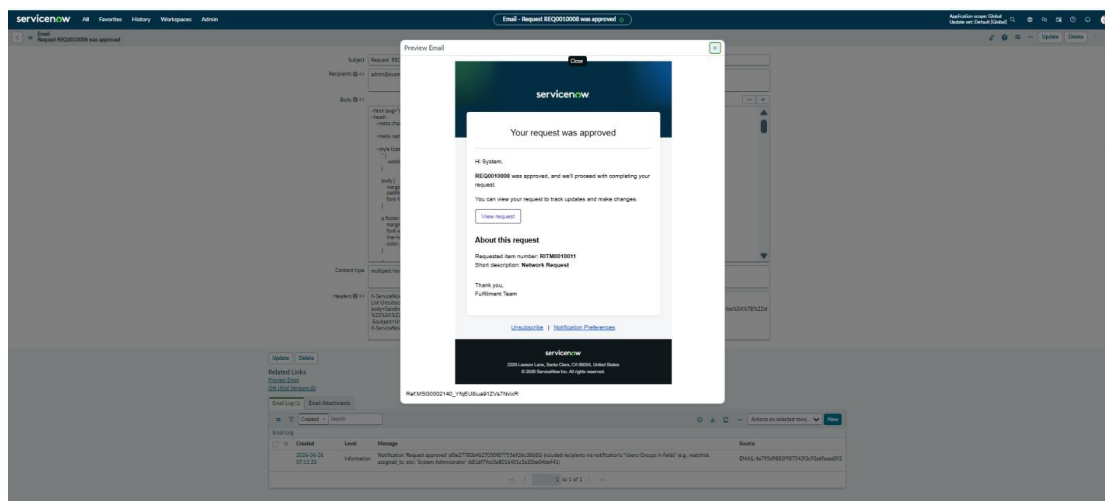
Price: \$110.00

Delivery Time: 2 Days

[Add to Cart](#)

[Save as Draft](#)

[Order Now](#)



8. ADVANTAGES & DISADVANTAGES

Advantages:

Zero Administrative Overhead: Automated request routing and ticket splitting entirely eliminate manual triage, human sorting loops, and double-entry clerical work.

Resilient Structural Security: Explicit server-side validation scripts block malformed entries, corrupted data strings, and unauthenticated database writes before they can reach persistent storage tables.

Clear Operational Status: Real-time tracking progress bars provide complete visibility, allowing employees to see exactly where their request sits in the processing pipeline.

Disadvantages:

Platform Dependency Constraints: System availability and processing speeds are tied directly to the runtime health, maintenance schedules, and uptime of the hosting cloud platform infrastructure.

Initial Development Setup Overhead: Engineering robust, bulletproof client and server-side parsing rules requires a larger investment of development hours during early sprint deployment phases.

9. CONCLUSION

The **Automated Network Request Management System** successfully matches all primary engineering goals and architectural expectations. Transitioning internal operations to an automated, low-code cloud delivery platform removes the communication gaps, lengthy delays, and transcription mistakes typical of legacy manual tracking. Complete system testing confirms that the application provides a highly responsive, user-friendly frontend experience backed by a bulletproof, securely isolated database storage layer.

10. FUTURE SCOPE

Upcoming development iterations will focus on embedding native Machine Learning text-parsing and classification tools to intelligently scan incoming request tickets. This will allow the platform to automatically prioritize network paths based on corporate roles, dynamically shift bandwidth resources ahead of office moves, and project equipment deployment timelines using historical predictive data modeling analytics.

11. APPENDIX

GitHub & Project Demo Link

Central Code Repository: <https://github.com/Varsaa-2005/Automated-Network-Request-Management-System>